

A DATA ANALYSIS WALKTHROUGH

Festive Season Sales: Who's Really Buying?

An exploratory data analysis of 11,000+ festive-season retail orders

Jobyar Ahmed | Python · Pandas · Seaborn

What is EDA, and why does it matter?

Exploratory Data Analysis is the process of examining data before drawing conclusions from it.

1

Understand the structure

See what the data actually contains, its shape, types, and gaps, before assuming anything about it.

2

Spot patterns early

Surface trends across customer groups, regions, and products that raw numbers hide.

3

Catch errors before they cost you

Missing values and bad records are found and fixed before they quietly skew a decision.

*Every model or business decision built on misunderstood data is unreliable from the start.
EDA is what turns a raw spreadsheet into something worth acting on.*

The Dataset

Retail orders placed during a major festive shopping season

11,251

raw rows collected

11,239

clean rows after removing nulls

15

columns of customer & order data

Fields covered

Gender · Age · Marital Status · Occupation · State & Zone · Product Category · Orders · Amount

Tools used

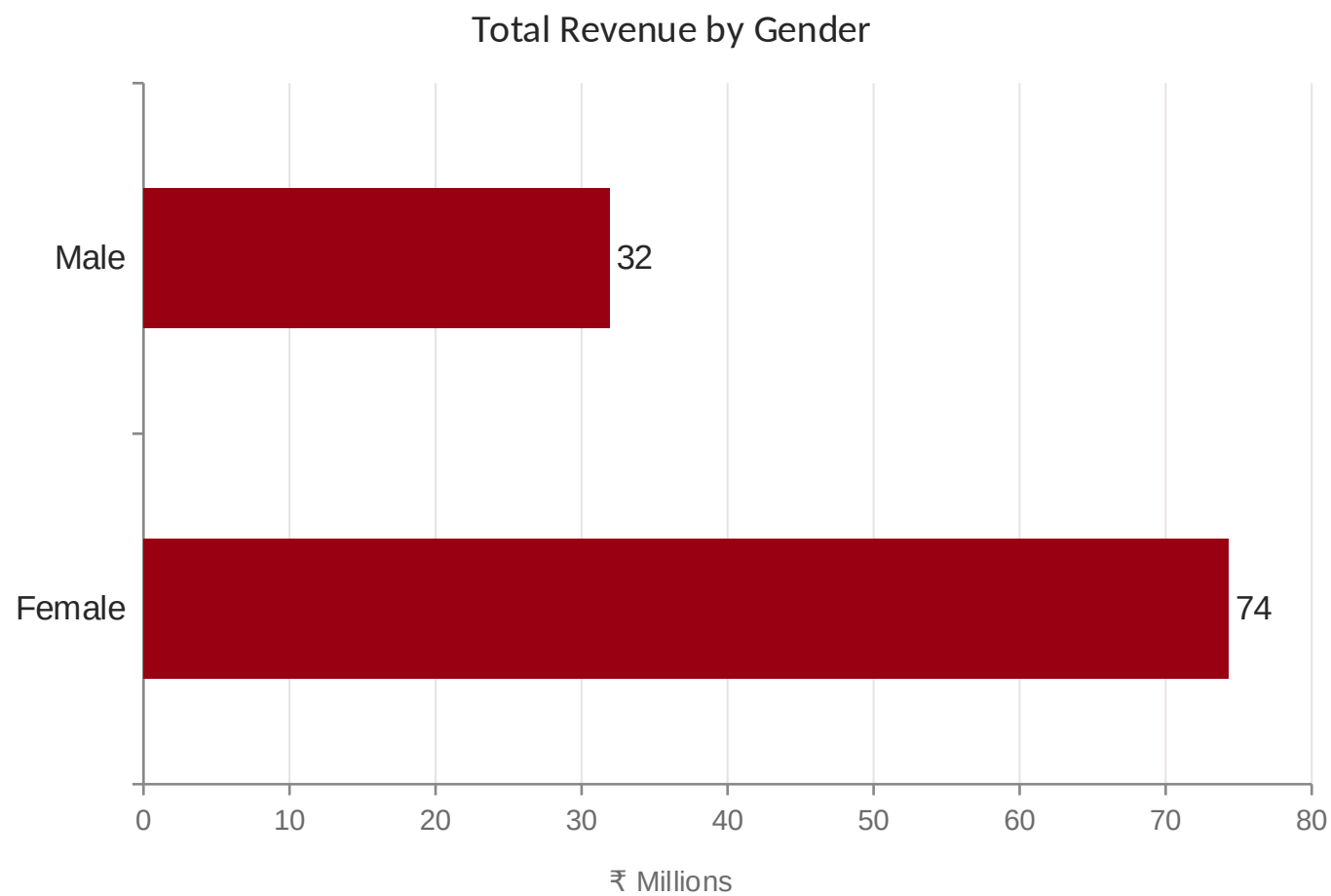
Python

Pandas & NumPy

Matplotlib

Seaborn

Who's Buying: Gender



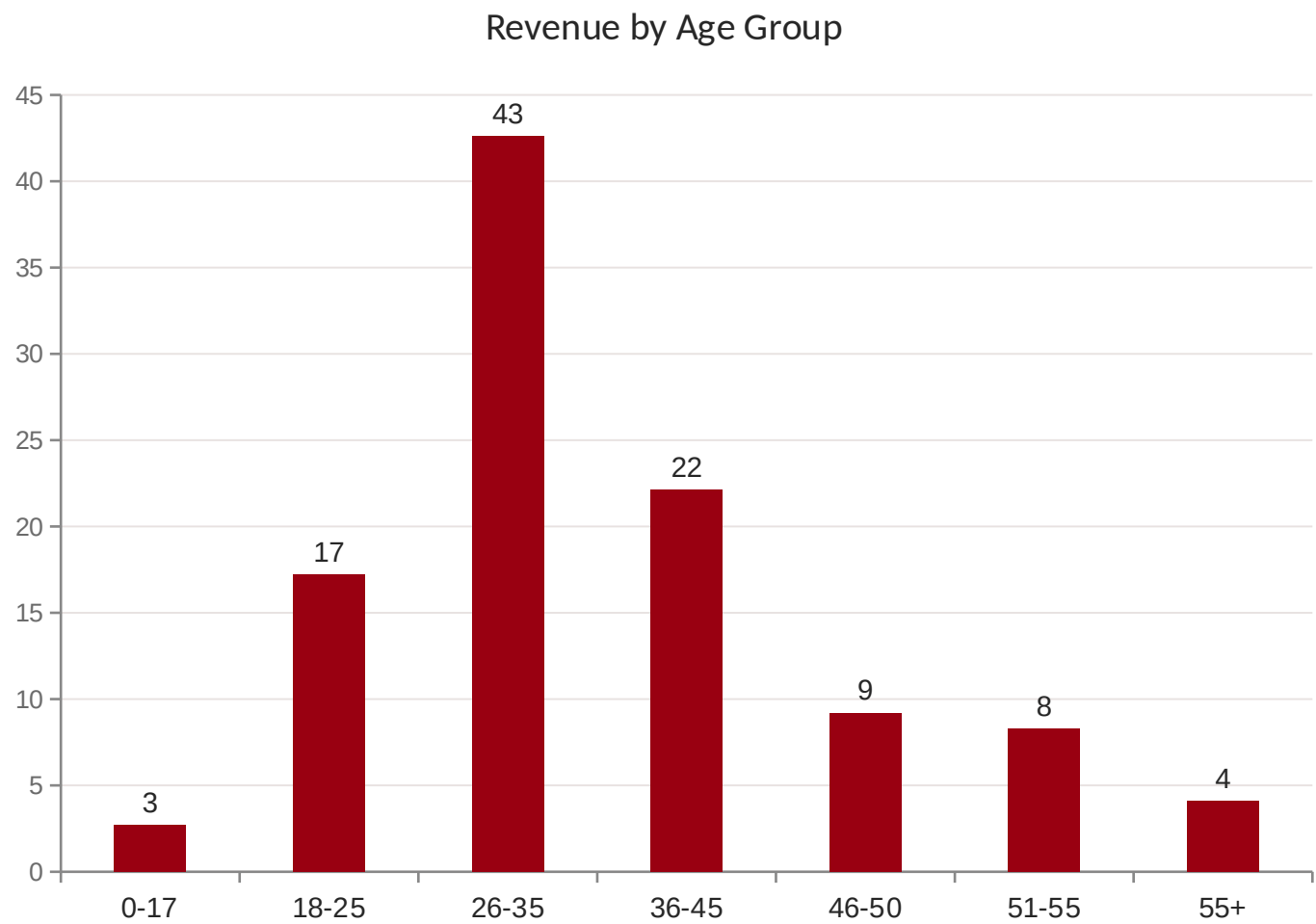
70%

of total festive-season revenue
came from female shoppers

7,832 vs 3,407

orders placed by women vs. men,
nearly 2.3x more orders overall

Who's Buying: Age & Marital Status



40%

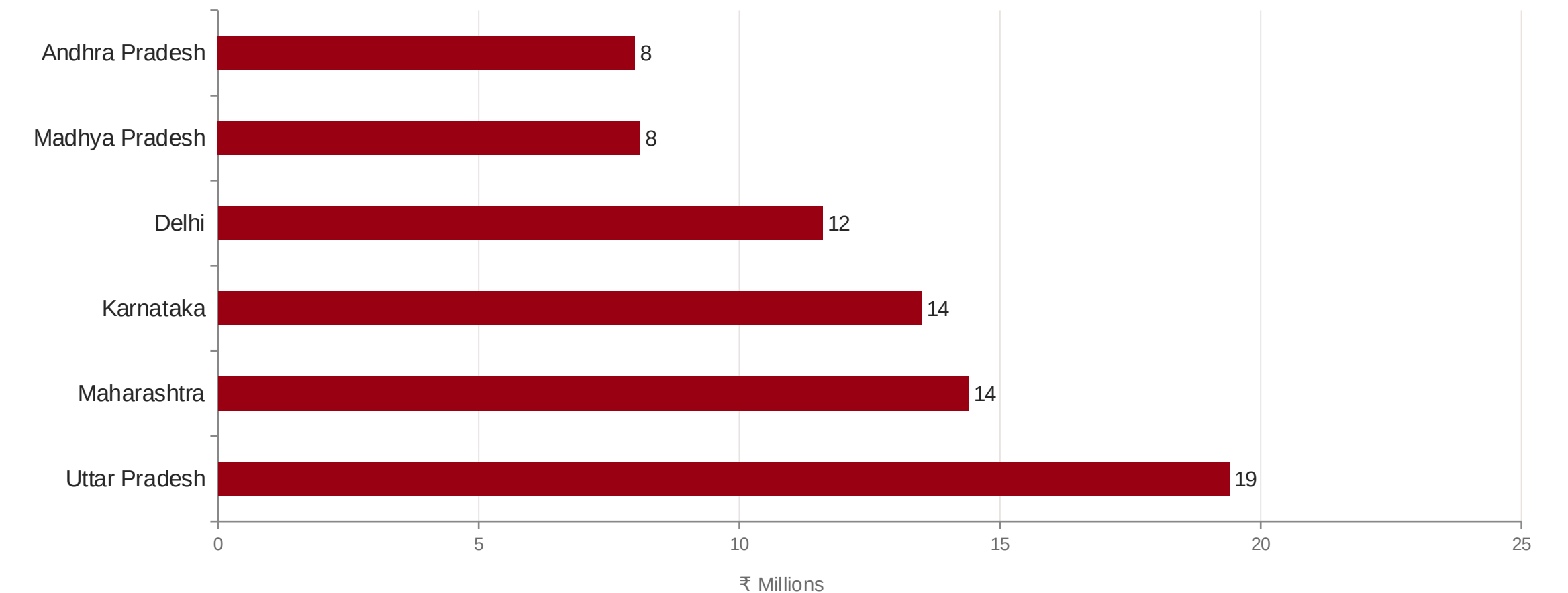
of all revenue comes from
shoppers aged 26–35 alone

Marital status

Within both married and unmarried groups, women consistently out-spend men, and by the widest margin among married shoppers (₹30.5M vs ₹13.6M).

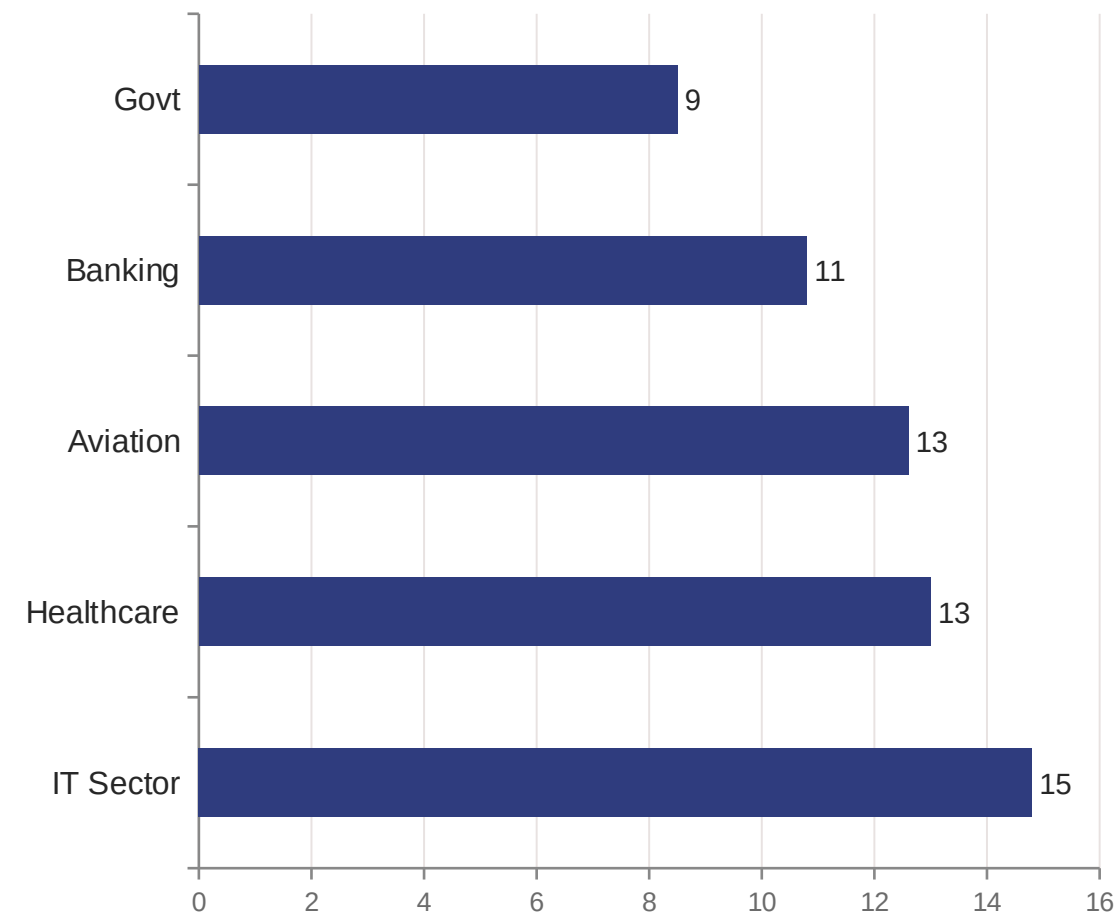
Where They're From: Top States

Top 3 states account for 44.5% of total revenue

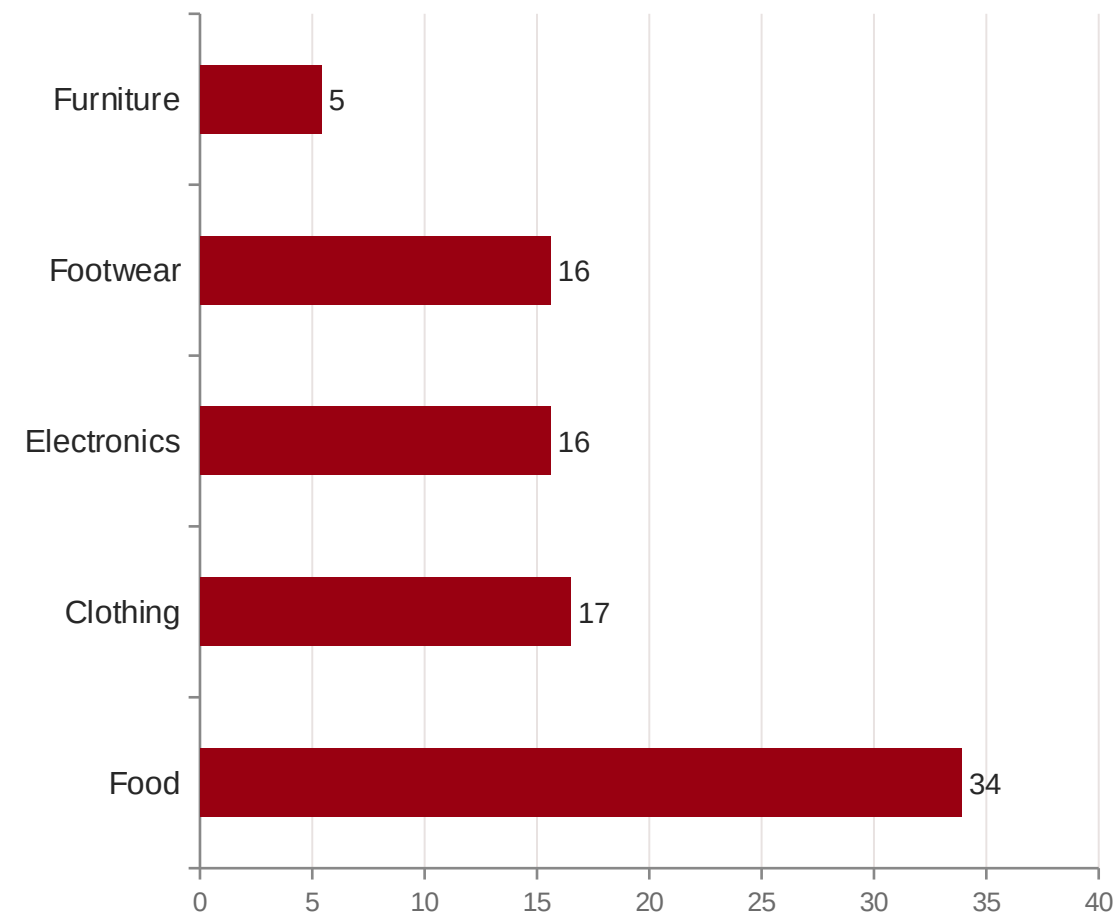


What They Do, What They Buy

Top occupations by spend



Top product categories



THE BOTTOM LINE

The core festive-season customer

- Married women
- Aged 26-35
- UP • Maharashtra • Karnataka
- IT • Healthcare • Aviation
- Buying Food, Clothing & Electronics

This is the segment worth designing festive-season campaigns around.

Small dataset, but a good reminder of how much a structured EDA process can reveal before any modeling even begins.